

FireBeetle ESP32 IOT Microcontroller (Supports Wi-Fi & Bluetooth) SKU: DFR0478



[FireBeetle ESP32 IOT Microcontroller](#)

Introduction

DFRobot FireBeetle series is the low-power consumption micro-controller intentionally designed for Internet of Things (IoT) projects. FireBeetle Board - ESP32 integrates a Dual-Core [ESP-WROOM-32 module](#), which supports MCU and Wi-Fi & Bluetooth dual-mode communication. The electric current is just 10 μ A in the deep-sleep mode. The main controller supports two power supply methods: USB and 3.7V external lithium battery. And both USB and external DC can charge the Lipo battery directly. FireBeetle Board-ESP32 has made a special hardware design for Arduino IDE. You can make a download without switching boot-mode manually. It supports Arduino, IDF (linux), micropython etc. Moreover, FireBeetle made a pin mapping for Arduino IDE. It can be configured with Dx transport, compatible with UNO and reduce the entry barrier.

The small size and high performance makes FireBeetle become idea solution for IOT low power consumption projects.

Please kindly note that developers are the most common users of [FireBeetle Board-ESP32](#) and not all of the peripherals have examples for reference, and there are still some bugs being found and fixed. For Arduino IDE, all GPIO function, I2C/SPI communication has been working. But the other functions are still under development. We recommend this board for experienced engineer.

Feature

- Compatible with products in the DFRobot FireBeetle serials
- Low-power consumption(the electricity current under ultra-low power is 10 μ A)
- Quick Response(the top frequency is 400KHz)
- Cost-effective
- Small size, convenient to install

Specification

- Working voltage: 3.3V

- Input voltage: 3.3V~5V
- Support electric current of low power consumption: 10 μ A
- Support maximum discharge current: 600mA
- Support maximum charge current: 500mA
- Support USB charging.
- Processor: Tensilica LX6 dual core processor (One for high speed connection; one for independent programming).
- Frequency: 240MHz
- SRAM : 520KB
- Flash : 16Mbit
- Wi-Fi standard : FCC/CE/TELEC/KCC
- Wi-Fi protocol: 802.11 b/g/n/d/e/I/k/r (802.11n, high speed can reach to 150 Mbps), converge A-MPDU and A-MSDU, supporting 0.4us protecting interval.
- Frequency range: 2.4~2.5 GHz
- Bluetooth protocol: Comply with BR/EDR/BLE standard of Bluetooth v4.2.
- Bluetooth audio: the current under low power consumption of CVSD and SBC is 10 μ A
- Working current: 80mA in average
- Frequency range: 2.4~2.5GHz
- Support one-key downloading.
- Support micropython.
- On-chip clock: 40MHz crystal and 32.768 KHz crystal.
- Digital I/O: 10 (default setting of arduino)
- Simulative input: 5(default setting of arduino)
- SPI: 1 (default setting of arduino)
- I2C: 1 (default setting of arduino)
- I2S: 1 (default setting of arduino)
- LED_BUILTIN : D9
- Interface: FireBeetle series compatible
- Working temperature: -40°C~+85°C
- Dimension: 24 × 53(mm)/0.94 x 2.09(inches)
- The dimension of mounting hole: inner diameter 3.1mm; outside diameter 6mm.

Board Overview

FireBeetle Board - ESP32 is not only compatible with ESP32 PinMap, but also make a special compatible with Arduino IDE PinMap. Dx(x=0,1,2,3...9)

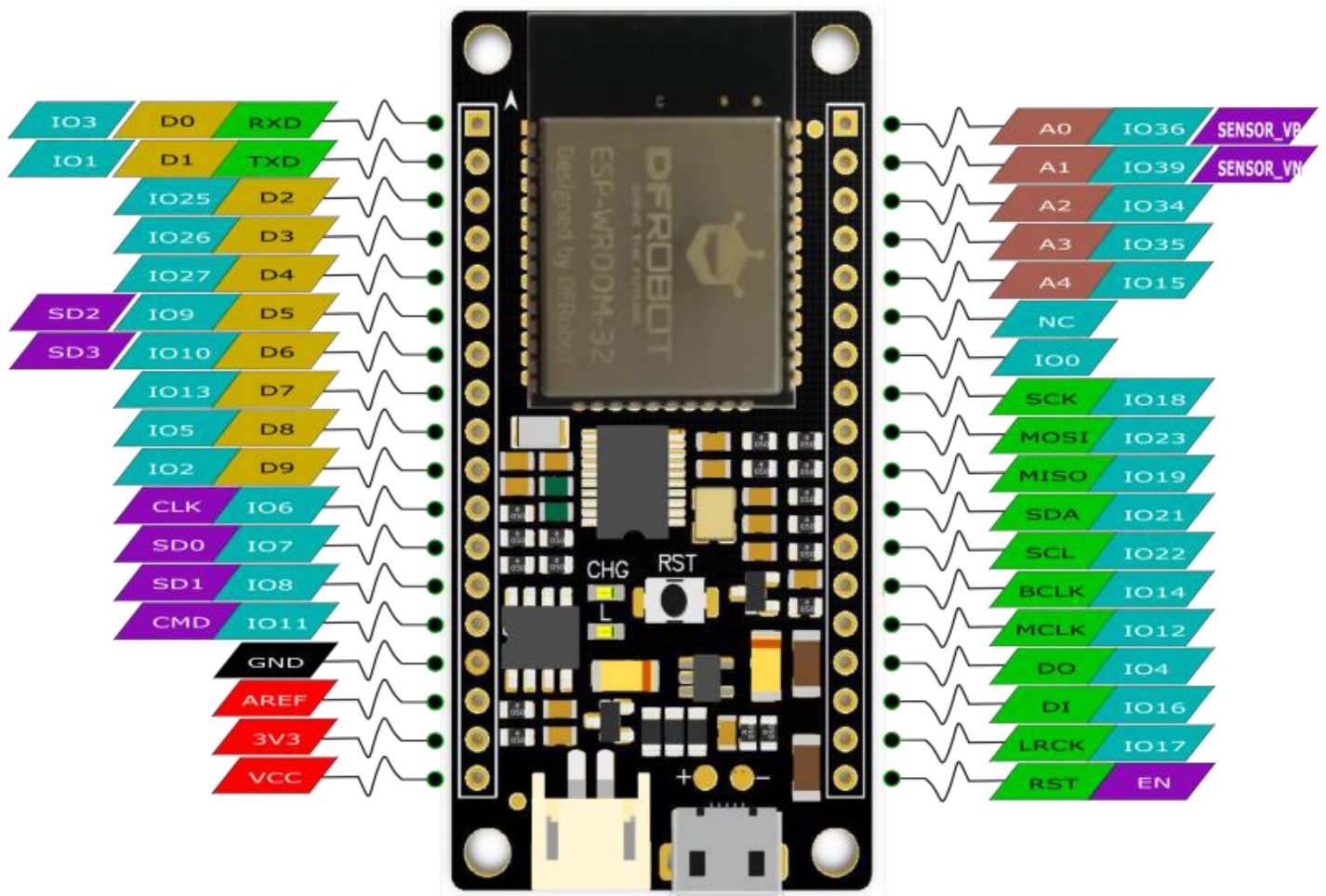


Fig1: FireBeetle Board-ESP32 PinOUT



Note: NC = Empty; VCC = VCC (5V under USB power supply, Around 3.7V under 3.7V lipo battery.power supply)

Tutorial

In this tutorial, we'll show you some basic operation with FireBeetle-ESP32 Microcontroller.

Requirements

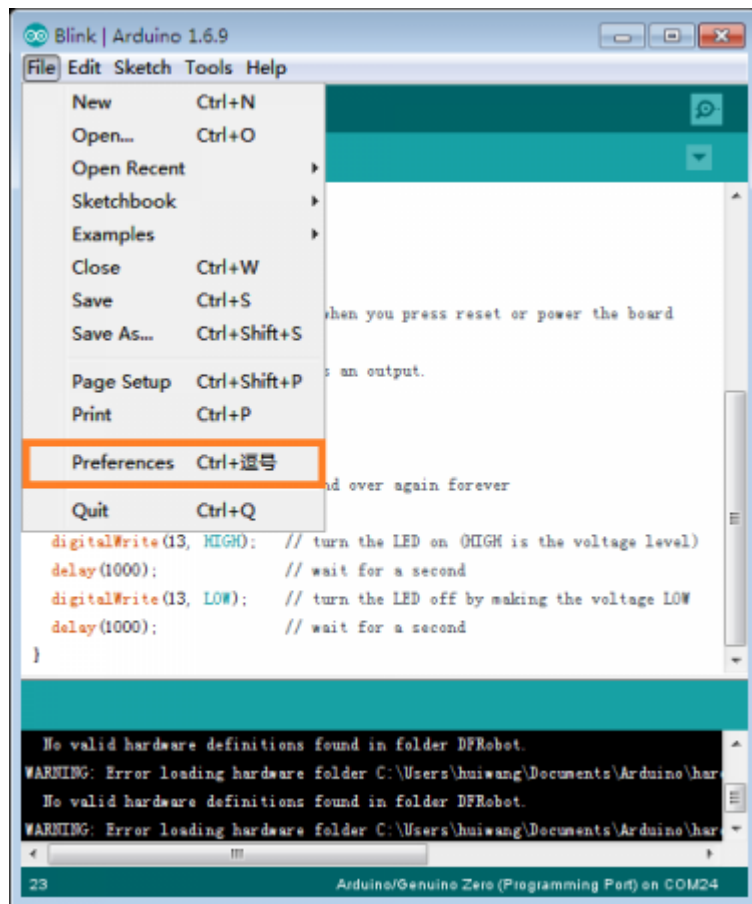
- **Hardware**
 - [FireBeetle ESP32 IOT Microcontroller](#) x 1
 - [Micro USB Cable](#) x1
- **Software**
 - Arduino IDE (newest), [Click to Download Arduino IDE from Arduino®](#)
 - [Download FireBeetle ESP32 Window Driver](#) **Note: Win8 + need disable the digital signature**

Setup Arduino IDE Development Environment

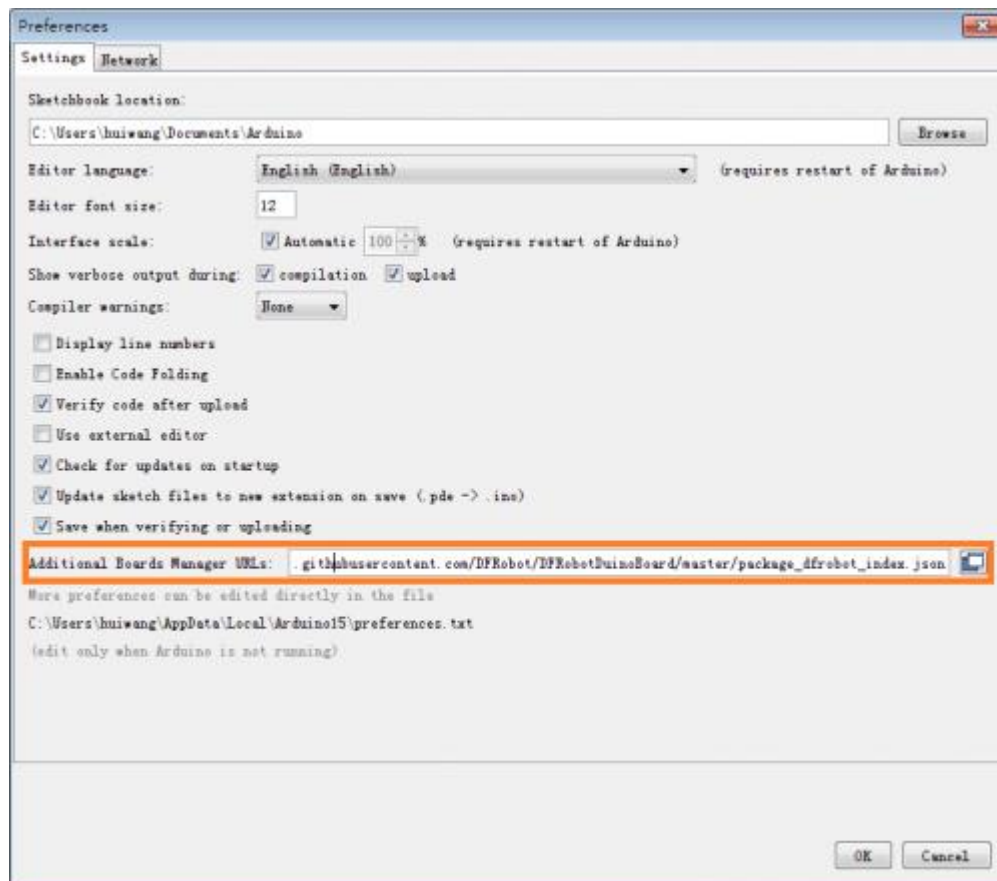
- Plug FireBeetle to your computer, install the driver manually.

- Add FireBeetle Board URL to Arduino IDE
- Open Arduino IDE, **File->Preferences**, find **Additional Boards Manager URLs**, copy the below link, and paste in the blank.

https://git.oschina.net/dfrobot/FireBeetle-ESP32/raw/master/package_esp32_index.json

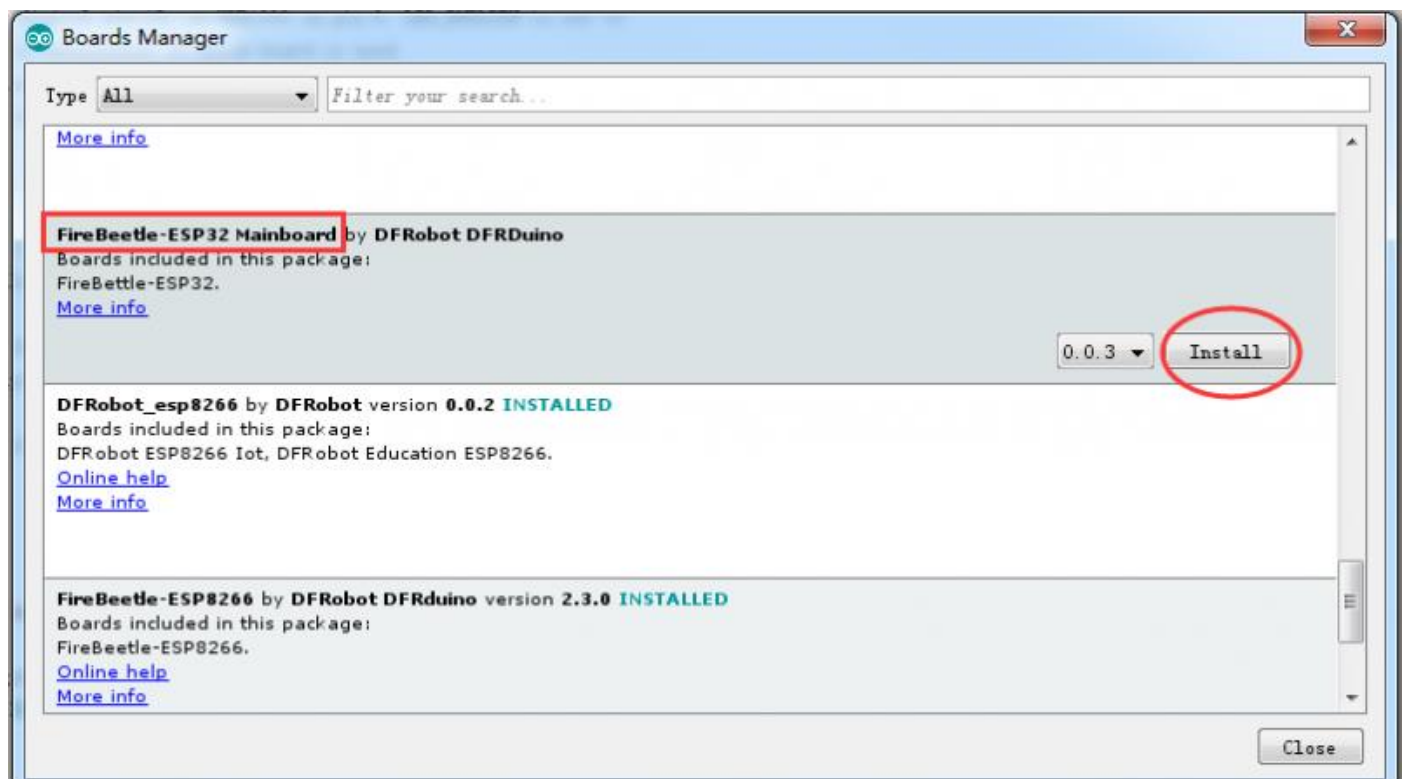


File->Preferences

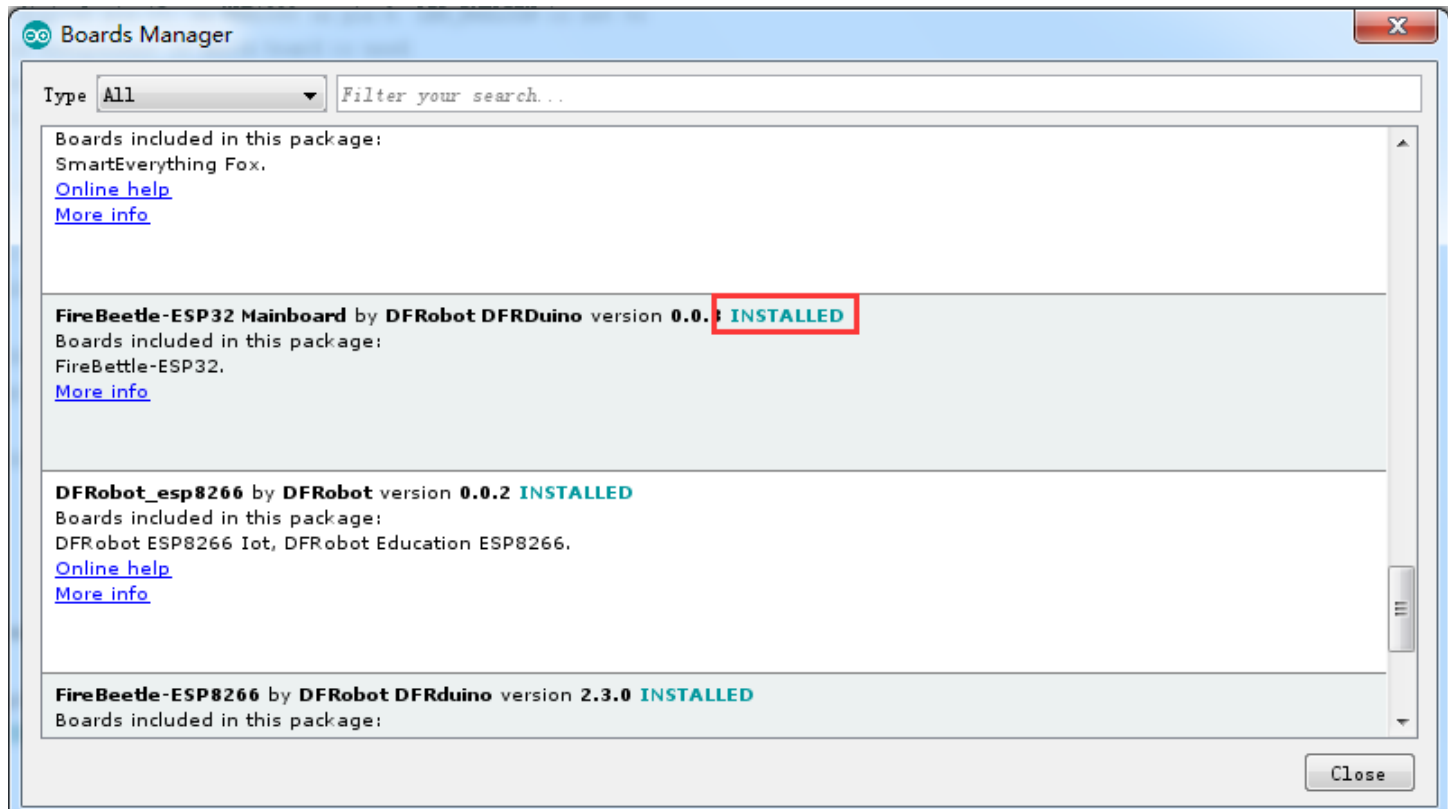


paste url here

- Click **OK**
- Open **Tools->Board->Boards Manager**, waiting automatic update. You'll find **FireBeetle-ESP32**



Now, the development environment has been installed, you can use it like a normal Arduino board.



Sample Code - Blink

The default LED for FireBeetle Board-ESP32 is D9, input following code:

```
// the setup function runs once when you press reset or power the board
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(LED_BUILTIN, OUTPUT);
}

// the loop function runs over and over again forever
void loop() {
  digitalWrite(LED_BUILTIN, HIGH);    // turn the LED on (HIGH is the voltage level)
  delay(1000);                       // wait for a second
  digitalWrite(LED_BUILTIN, LOW);     // turn the LED off by making the voltage LOW
  delay(1000);                       // wait for a second
}
```

For more code and instruction, please refer to [FireBeetle User Manual](#)

FAQ

Q1. Some general Arduino Problems/FAQ/Tips

A. Click [the topic link](#) on DFRobot Forum.

For any questions, advice or cool ideas to share, please visit the [DFRobot Forum](#).

More

- [FireBeetle Arduino Bluetooth Examples](#)
- [FireBeetle Schematic](#)